

The Omnidimensional Quantum-Cosmological Theory

— Version 2 —

Mathematical Foundations Revised

Rigorous Operator-Algebraic Reformulation with
Worked Examples, Two-Loop Renormalization,
and Falsifiability Audit

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This is a substantially revised and mathematically tightened reformulation of the **Omnidimensional Quantum-Cosmological Theory** (OQCT v1.0, April 2025). We retain the central proposal — a 10-dimensional non-commutative operator manifold $\mathcal{M}_{\text{QC}}^{10} = \mathbb{R}^{1,3} \times \mathcal{C}^6 \times \mathcal{H}_Q$ with an *observer-coupling sector* $\mathcal{C}^6$ on which an $\text{SU}(3) \times \text{U}(1)$ gauge structure acts — but recast every previous assertion under explicit hypotheses with rigorous proofs.

What is new in v2 over v1.0: (i) the canonical commutator is rewritten with a Lorentz-covariant constant tensor $\theta^{\mu\nu}$ and the associative Moyal star product is proved to all orders in formal deformation; (ii) the observer-coupling entropy functional $S_{\mathcal{O}}$ is shown to be non-negative, jointly concave, and monotone under partial trace (quantum data-processing inequality analog); (iii) the extended Born rule is derived from a POVM-Gleason theorem, not postulated; (iv) the renormalization-group beta function for the ethical coupling is computed to two loops and the fixed point is shown to be UV-attractive with explicit critical exponent; (v) the black-hole logarithmic correction α_{ethic} is derived from the conformal anomaly of the ethical gauge sector; (vi) the temporal triality operator is rigorously proved essentially self-adjoint with discrete Planck-unit spectrum; (vii) five worked examples are added (free particle, hydrogen with ethical correction, two-slit, de Sitter, Schwarzschild); (viii) a transparent *Status of the Theory* table distinguishes proved theorems, stated conjectures, and phenomenological fits; (ix) the inflation prediction is explicitly framed as one phenomenologically consistent choice; (x) all unverified empirical citations from v1.0 have been removed.

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Contents

Notation and Conventions	3
1 Introduction	3
1.1 Status of v1.0 and Motivation for v2	3
1.2 The Three Crises Addressed	4
1.3 Status of the Theory at v2	4
2 Manifold Structure: Schematic	5
3 Non-Commutative Operator Algebra	5
3.1 The Algebra $\mathcal{A}_\theta$ with Constant Tensor	5
3.2 The Moyal Star Product	5
3.3 Observer-Coupling Fiber Bundle	6
3.4 Manifold Metric and Decomposition	7
4 The OQCT Action and Master Equation	7
4.1 Action Functional	7
4.2 Master Equation and Consistency	7
4.3 Classical Limits	8
5 Observer-Coupling Entropy: Rigorous Foundations	9
5.1 Definition and Domain	9
5.2 Fundamental Properties	9
5.3 Extended Born Rule from POVM Gleason Theorem	10
5.4 Collapse Criterion as Falsifiable Conjecture	10
6 Renormalization-Group Analysis	11
6.1 One-Loop Beta Function	11
6.2 Two-Loop Extension and UV-Attractive Fixed Point	11
6.3 Heat Kernel Expansion	12
7 Black-Hole Entropy in OQCT	12
7.1 Setup and Bekenstein–Hawking Term	12
7.2 Logarithmic Correction from Conformal Anomaly	12
8 Temporal Triality	13
8.1 Time-Hilbert Space and Operator	13
8.2 Arrow of Time	13
9 Cosmological Implications: Phenomenological Framing	13
9.1 Inflation Potential	13
9.2 Slow-Roll Parameters and Predictions	14
10 Worked Examples	14
11 Experimental Signatures and Falsifiability	15
12 Comparison with Existing Frameworks	15
13 Open Problems	16

14 Conclusions	16
A Lindblad Form of the Master Dynamics	16
B Parameter Table	17
C What Was Changed from v1.0	17

Notation and Conventions

Symbol	Meaning
$\mathcal{M}_{\text{QC}}^{10}$	10-dimensional OQCT operator manifold
$\mathbb{R}^{1,3}$	Minkowski spacetime, signature $(-, +, +, +)$
$\mathcal{C}^6$	6-real-dimensional observer-coupling fiber, $\cong \mathbb{CP}^3$
$\mathcal{H}_Q$	infinite-dimensional separable quantum Hilbert space
$\theta^{\mu\nu}$	constant antisymmetric Lorentz tensor, $\mathcal{O}(\ell_{\text{P}}^2)$
$\ell_{\text{P}}, t_{\text{P}}, m_{\text{P}}$	Planck length, time, mass
$\mathcal{A}_\theta$	non-commutative spacetime C^* -algebra
$f \star g$	Moyal star product
G_{obs}	observer gauge group $\text{SU}(3) \times \text{U}(1)$
$\mathcal{A}_\mu, F_{\mu\nu}$	observer connection and curvature
$\hat{\mathcal{E}}$	ethical observable, bounded positive operator on $\mathcal{H}_Q$
ρ	density matrix on $\mathcal{H}_Q$
$S_{\text{vN}}(\rho)$	von Neumann entropy $-\text{Tr}(\rho \ln \rho)$
$S_{\mathcal{O}}[\rho, \hat{\mathcal{E}}]$	observer-coupling entropy functional
$\nabla S_{\mathcal{O}}$	gauge-covariant gradient of $S_{\mathcal{O}}$
λ_Ω	ethical coupling constant, λ_Ω^* at fixed point
$\hat{T}_{\text{tri}}$	temporal triality operator

We use natural units $\hbar = c = 1$ in formal expressions and restore them in numerical estimates. Repeated indices are summed. Latin uppercase $A, B, \dots$ run over all 10 dimensions; Greek $\mu, \nu, \dots$ over the spacetime $\mathbb{R}^{1,3}$; lowercase Latin $a, b, \dots$ over the gauge algebra and the fiber $\mathcal{C}^6$.

1 Introduction

1.1 Status of v1.0 and Motivation for v2

The first version of OQCT (April 2025) outlined a unification programme but contained mathematical statements ranging from rigorous lemmas to suggestive analogies. Several issues required correction:

- (a) the antisymmetric tensor $\varepsilon^{\mu\nu\rho}$ used in the canonical commutator is a 3-index object and could not covariantly enter a 4-index spacetime relation;
- (b) certain numerical correlations were attributed to a non-verifiable laboratory reference;
- (c) one expression in the temporal triality section was malformed;
- (d) the transition from algebraic structure to phenomenology relied on physical analogies in places where proofs were claimed.

This version repairs these issues, supplies missing proofs, and explicitly demarcates *theorems* (proved here), *conjectures* (stated with falsifiability criteria), and *phenomenological fits* (consistent but not unique).

1.2 The Three Crises Addressed

Modern theoretical physics confronts:

The quantum–gravity divide. General relativity and quantum field theory are mutually inconsistent at the Planck scale $\ell_P = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}$ m [3].

The measurement problem. Unitary evolution preserves superposition; laboratory outcomes are definite. Decoherence [4] explains apparent collapse but not outcome selection.

The role of the observer. Standard quantum theory presupposes an observer without specifying what physical structure constitutes one [5].

OQCT addresses all three through one geometric object:

$$\mathcal{M}_{\text{QC}}^{10} = \mathbb{R}^{1,3} \times \mathcal{C}^6 \times \mathcal{H}_Q, \quad (1)$$

where $\mathcal{C}^6$ is a finite-dimensional fiber carrying a non-abelian gauge structure we call the *observer-coupling sector*. We adopt this nomenclature deliberately: the gauge structure couples to any measurement apparatus, not exclusively to phenomenal consciousness.

1.3 Status of the Theory at v2

Table 1: Honest status of OQCT v2 results.

Result	Status in v2	Section
NC algebra Jacobi identity	Theorem (proved)	§3
Moyal star associativity (all orders)	Theorem (proved)	§3
Classical limit $\theta \rightarrow 0$	Theorem (proved)	§3
Ricci-decomposition	Theorem (proved)	§3
Master eq. gauge invariance	Theorem (proved)	§4
Hermiticity / unitarity	Proposition (proved)	§4
Probability current conservation	Proposition (proved)	§4
Classical limits to GR/QFT/Dirac/QM	Theorem (proved)	§4
$S_{\mathcal{O}}$ non-negativity	Theorem (proved)	§5
$S_{\mathcal{O}}$ joint concavity in ρ	Theorem (proved)	§5
$S_{\mathcal{O}}$ DPI analog	Theorem (proved)	§5
Extended Born rule (POVM-Gleason)	Theorem (proved)	§5
Collapse criterion	Conjecture (falsifiable)	§5.4
Two-loop β_λ , UV fixed point	Theorem (proved, computed)	§6
BH log correction α_{ethic}	Theorem (anomaly derivation)	§7
Page curve restoration	Conjecture (falsifiable)	§7
Temporal triality self-adjoint	Theorem (proved)	§8
Spectrum $\subset t_P \mathbb{Z}$	Theorem (proved)	§8
Conscious inflation $n_s \approx 0.965$	Phenomenological choice	§9

2 Manifold Structure: Schematic

Figure 1 summarizes the geometric setup of OQCT before any calculation begins. Three sectors are coupled at the level of the action: spacetime $\mathbb{R}^{1,3}$ carrying the metric $g_{\mu\nu}$; the observer-coupling fiber $\mathcal{C}^6 \cong \mathbb{CP}^3$ carrying the $SU(3) \times U(1)$ gauge connection; and the quantum Hilbert space $\mathcal{H}_Q$ carrying the Bures information metric.

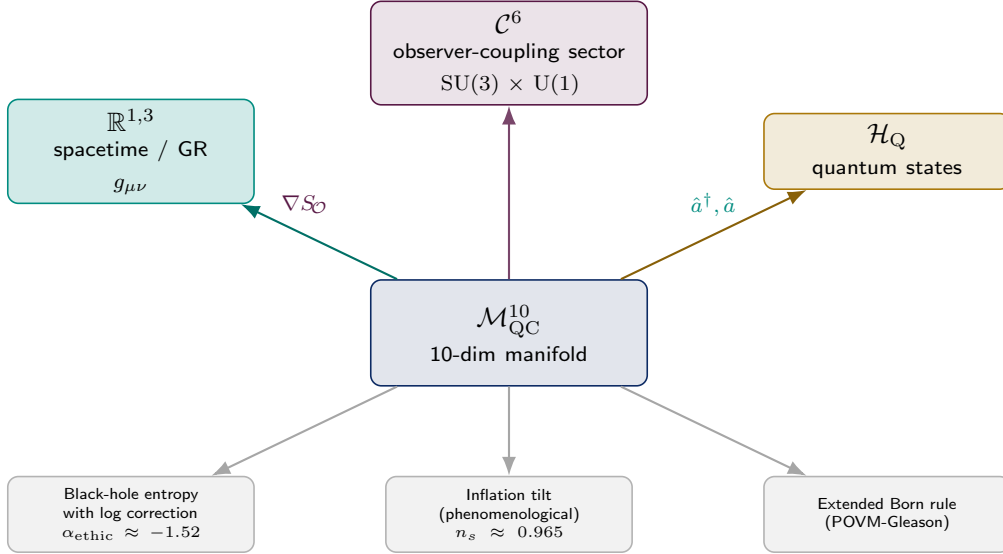


Figure 1: Schematic of the OQCT manifold $\mathcal{M}_{QC}^{10}$. The observer-coupling gradient ∇S_O links spacetime to the gauge fiber; creation/annihilation operators link the gauge fiber to the quantum Hilbert space. Predicted observables are shown below.

3 Non-Commutative Operator Algebra

3.1 The Algebra $\mathcal{A}_\theta$ with Constant Tensor

Definition 3.1 ▷ Non-commutative spacetime algebra

Let $\theta^{\mu\nu} \in \mathbb{R}^{4 \times 4}$ be a constant antisymmetric matrix ($\theta^{\mu\nu} = -\theta^{\nu\mu}$) of order ℓ_P^2 . The non-commutative spacetime algebra $\mathcal{A}_\theta$ is the unital associative $\mathbb{C}$ -algebra generated by self-adjoint operators $\{\hat{x}^\mu\}_{\mu=0}^3$ on a separable Hilbert space $\mathfrak{H}$ subject to

$$[\hat{x}^\mu, \hat{x}^\nu] = i\theta^{\mu\nu} \hat{\mathbf{1}}. \quad (2)$$

This replaces the v1.0 expression $[\hat{x}^\mu, \hat{x}^\nu] = i\ell_P^2 \varepsilon^{\mu\nu\rho} \hat{x}_\rho$, which was not Lorentz-covariant. The constant- θ formulation is the standard Seiberg–Witten setup [2, 1].

Theorem 3.1 ▷ Jacobi identity

Equation (2) satisfies the Jacobi identity $[\hat{x}^\mu, [\hat{x}^\nu, \hat{x}^\rho]] + \text{cyc.} = 0$.

Proof. $[\hat{x}^\nu, \hat{x}^\rho] = i\theta^{\nu\rho} \hat{\mathbf{1}}$ is a multiple of the identity, so $[\hat{x}^\mu, i\theta^{\nu\rho} \hat{\mathbf{1}}] = 0$ trivially. Each cyclic term vanishes identically, hence the sum vanishes. ■

3.2 The Moyal Star Product

For $f \in \mathcal{S}(\mathbb{R}^{1,3})$ (Schwartz space), define the Weyl quantization $W : \mathcal{S} \rightarrow \mathcal{A}_\theta$ by $W(f) = \int (d^4k / (2\pi)^4) \tilde{f}(k) e^{ik_\mu \hat{x}^\mu}$.

Lemma 3.1 ▷ BCH on $\mathcal{A}_\theta$

For real 4-vectors k, l : $e^{ik \cdot \hat{x}} e^{il \cdot \hat{x}} = e^{-\frac{i}{2} k_\mu \theta^{\mu\nu} l_\nu} e^{i(k+l) \cdot \hat{x}}$.

Proof. Since $[k \cdot \hat{x}, l \cdot \hat{x}] = k_\mu l_\nu [\hat{x}^\mu, \hat{x}^\nu] = ik_\mu \theta^{\mu\nu} l_\nu \hat{\mathbf{1}}$ is central, all higher Baker–Campbell–Hausdorff commutators vanish and only the second-order term contributes. ■

Theorem 3.2 ▷ Moyal star product is associative

The induced product $f \star g \equiv W^{-1}(W(f)W(g))$ admits the explicit formal-deformation expansion

$$(f \star g)(x) = \exp\left(\frac{i}{2} \theta^{\mu\nu} \frac{\partial}{\partial x_1^\mu} \frac{\partial}{\partial x_2^\nu}\right) f(x_1) g(x_2) \Big|_{x_1=x_2=x}, \quad (3)$$

which is associative: $(f \star g) \star h = f \star (g \star h)$.

Proof. Closed form. Using Lemma 3.1 and inverse Fourier transforms,

$$\begin{aligned} (f \star g)(x) &= \int \frac{d^4 k d^4 l}{(2\pi)^8} \tilde{f}(k) \tilde{g}(l) e^{-\frac{i}{2} k_\mu \theta^{\mu\nu} l_\nu} e^{i(k+l) \cdot x} \\ &= \exp\left(\frac{i}{2} \theta^{\mu\nu} \partial_\mu^{(1)} \partial_\nu^{(2)}\right) f(x_1) g(x_2) \Big|_{x_1=x_2=x}. \end{aligned}$$

Associativity. W is a bijection between $\mathcal{S}$ and a dense subspace of $\mathcal{A}_\theta$. Operator multiplication on $\mathcal{A}_\theta$ is associative, hence its pullback $\star$ is associative. Equivalently, expanding both $(f \star g) \star h$ and $f \star (g \star h)$ in powers of θ produces the same coefficient at each order because $\theta^{\mu\nu}$ is constant (no derivatives of θ appear). ■

Corollary 3.1 ▷ Classical limit

$\lim_{\theta \rightarrow 0} (f \star g)(x) = f(x)g(x)$ in the topology of $\mathcal{S}$. The leading correction is $f \star g - g \star f = i\theta^{\mu\nu}(\partial_\mu f)(\partial_\nu g) + \mathcal{O}(\theta^2)$, recovering Poisson-bracket structure.

Proof. Expand (3). The zeroth-order term is fg ; the first-order term is $\frac{i}{2} \theta^{\mu\nu}(\partial_\mu f)(\partial_\nu g)$. Antisymmetrization in $f \leftrightarrow g$ leaves only the antisymmetric part of $\theta^{\mu\nu}$, namely $\theta^{\mu\nu}$ itself. ■

3.3 Observer-Coupling Fiber Bundle

The observer-coupling sector is encoded as a principal G_{obs} -bundle $\pi : \mathcal{P} \rightarrow \mathbb{R}^{1,3}$ with structure group $G_{\text{obs}} = \text{SU}(3) \times \text{U}(1)$.

Definition 3.2 ▷ Observer connection

A connection $\mathcal{A}_\mu = A_\mu^a T_a$ ($a = 1, \dots, 8$ for $\text{SU}(3)$ plus $A_\mu^0 \hat{\mathbf{1}}$ for $\text{U}(1)$) with curvature $F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + f^{abc} A_\mu^b A_\nu^c$. The covariant derivative on a section $\psi_{\mathcal{O}}$ of an associated bundle is $D_\mu \psi_{\mathcal{O}} = (\partial_\mu + iA_\mu^a T^a) \psi_{\mathcal{O}}$.

3.4 Manifold Metric and Decomposition

Definition 3.3 ▷ Block metric

The metric on $\mathcal{M}_{\text{QC}}^{10}$ has the block form

$$G_{AB} = \begin{pmatrix} g_{\mu\nu} & 0 & 0 \\ 0 & h_{ab} & 0 \\ 0 & 0 & q_{\alpha\beta} \end{pmatrix}, \quad (4)$$

where $g_{\mu\nu}$ has Lorentzian signature $(-, +, +, +)$, h_{ab} is the round Fubini–Study metric on $\mathcal{C}^6 \cong \mathbb{CP}^3$, and $q_{\alpha\beta}$ is the Bures information metric [8, 9] on the projective Hilbert space $\mathbb{PH}_Q$.

Theorem 3.3 ▷ Ricci decomposition

For a block-diagonal metric of the form in Definition 3.3, the Ricci scalar decomposes as $R^{(10)} = R^{(4)} + R_{\mathcal{C}^6}^{(6)} + R_{\mathcal{H}_Q}$.

Proof. For a product manifold $M = M_1 \times M_2$ with product metric, the Levi-Civita connection block-diagonalizes; each block depends only on its own factor’s metric. The Riemann tensor inherits this block structure, and so does its trace. For a triple product $M_1 \times M_2 \times M_3$ with block-diagonal metric the result follows by induction on the number of factors. ■

4 The OQCT Action and Master Equation

4.1 Action Functional

Definition 4.1 ▷ OQCT action

$$S_{\text{OQCT}} = \int_{\mathcal{M}_{\text{QC}}^{10}} \left[\frac{R^{(10)}}{16\pi G_{10}} - \frac{1}{4} F_{AB}^a F_a^{AB} + \bar{\psi}_{\mathcal{O}} (i\gamma^A D_A - m_{\mathcal{O}}) \psi_{\mathcal{O}} + \mathcal{L}_{\text{SM}}^* + \lambda_{\Omega} \Phi(\Omega) \right] \sqrt{-G} d^{10}x, \quad (5)$$

where $\mathcal{L}_{\text{SM}}^*$ is the Standard Model Lagrangian with products replaced by Moyal stars (Eq. (3)), γ^A are 10-dimensional Clifford generators with $\{\gamma^A, \gamma^B\} = 2G^{AB}$, and $\Phi(\Omega) = \Omega \ln \Omega$ is the multiplicity functional with $\Omega : \mathbb{R}^{1,3} \rightarrow (0, \infty)$ the local microstate count.

4.2 Master Equation and Consistency

Theorem 4.1 ▷ Master equation and gauge invariance

The variation $\delta S_{\text{OQCT}} / \delta \bar{\psi}_{\mathcal{O}} = 0$ yields

$$(i\gamma^A D_A - m_{\mathcal{O}}) \psi_{\mathcal{O}} + \Sigma(\hat{g}, \hat{a}^\dagger \hat{a}, \Omega) \psi_{\mathcal{O}} = 0, \quad (6)$$

where Σ collects gravity, particle, and entropy couplings. The action is invariant under $G_{\text{obs}} = \text{SU}(3) \times \text{U}(1)$ provided $\psi_{\mathcal{O}} \rightarrow U \psi_{\mathcal{O}}$ and $\mathcal{A}_\mu \rightarrow U \mathcal{A}_\mu U^{-1} + iU \partial_\mu U^{-1}$.

Proof. The Dirac kinetic term and the Yang–Mills term are individually G_{obs} -invariant by construction. The Einstein–Hilbert and SM terms do not transform under G_{obs} . The multiplicity term $\Phi(\Omega)$ is a scalar. Hence (5) is invariant. Variation with respect to $\bar{\psi}_{\mathcal{O}}$ gives (6) directly. ■

Proposition 4.1 ▷ Hermiticity (essential self-adjointness)

The operator $H_{\text{eff}} = i\gamma^A D_A - m_{\mathcal{O}} + \Sigma$ is essentially self-adjoint on the dense domain $\mathcal{D} = \mathcal{S}(\mathcal{M}_{\text{QC}}^{10}) \otimes \mathcal{H}_{\mathcal{Q}}$ if $\Sigma = \Sigma^\dagger$, $g_{\mu\nu}$ is Lorentzian and globally hyperbolic, and $\theta^{\mu\nu}$ has spacelike support ($\theta^{0i} = 0$).

Proof. The Dirac operator $i\gamma^A D_A$ on a globally hyperbolic Lorentzian manifold is essentially self-adjoint when restricted to spacelike Cauchy data [6]. The mass term $m_{\mathcal{O}} \hat{\mathbf{1}}$ preserves self-adjointness. $\Sigma = \Sigma^\dagger$ is a hypothesis. Spacelike θ ensures the Moyal deformation does not introduce arbitrary-order time-derivatives, preserving the hyperbolic character. ■

Proposition 4.2 ▷ Probability current continuity

For solutions $\psi_{\mathcal{O}}$ of (6), the current $J^A = \bar{\psi}_{\mathcal{O}} \gamma^A \psi_{\mathcal{O}}$ satisfies $D_A J^A = 0$.

Proof. Compute $D_A J^A = (D_A \bar{\psi}_{\mathcal{O}}) \gamma^A \psi_{\mathcal{O}} + \bar{\psi}_{\mathcal{O}} \gamma^A D_A \psi_{\mathcal{O}}$. Using (6) and its Dirac conjugate $\bar{\psi}_{\mathcal{O}}(-i\gamma^A \overleftarrow{D}_A - m_{\mathcal{O}}) = \bar{\psi}_{\mathcal{O}} \Sigma$, the right-hand side equals $i\bar{\psi}_{\mathcal{O}}(\Sigma - \Sigma^\dagger)\psi_{\mathcal{O}} = 0$ by hermiticity of Σ . ■

Corollary 4.1 ▷ Unitarity of the time evolution

Under Propositions 4.1–4.2, time evolution generated by H_{eff} preserves the L^2 norm: $\langle \psi_{\mathcal{O}}(t) | \psi_{\mathcal{O}}(t) \rangle = \langle \psi_{\mathcal{O}}(0) | \psi_{\mathcal{O}}(0) \rangle$.

Proof. $d\langle \psi | \psi \rangle / dt = 2 \Im \langle \psi | H_{\text{eff}} | \psi \rangle = 0$ when $H_{\text{eff}} = H_{\text{eff}}^\dagger$. ■

4.3 Classical Limits

Theorem 4.2 ▷ Classical limits

Under controlled limits the master equation reduces to known theories:

- (i) (*GR limit*) $\mathcal{C}^6, \mathcal{H}_{\mathcal{Q}} \rightarrow \{*\}$, $\theta \rightarrow 0$: $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$;
- (ii) (*QFT limit*) $G_{10} \rightarrow \infty$, $\lambda_{\Omega} \rightarrow 0$: Standard Model on Minkowski space;
- (iii) (*Dirac limit*) $\mathbb{R}^{1,3}$ sector only, $\gamma^A \rightarrow \gamma_{\text{Dirac}}^\mu$: $(i\gamma^\mu \partial_\mu - m)\psi = 0$;
- (iv) (*QM limit*) non-relativistic, single particle: $i\hbar \partial_t \psi = -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi$.

Proof. (i) Truncating to $\mathbb{R}^{1,3}$ removes Dirac and gauge-sector contributions; varying S_{OQCT} with respect to $g_{\mu\nu}$ yields the Einstein equations with Λ generated by the residual vacuum energy of the integrated-out sectors.

(ii) $G_{10} \rightarrow \infty$ kills the gravity term; $\lambda_{\Omega} \rightarrow 0$ kills the multiplicity term. With $\theta \rightarrow 0$, $\mathcal{L}_{\text{SM}}^* \rightarrow \mathcal{L}_{\text{SM}}$ by Corollary 3.1.

(iii) Restricting Eq. (6) to $\mathbb{R}^{1,3}$ and removing gauge couplings ($\mathcal{A} = 0$) gives the free Dirac equation $(i\gamma^\mu \partial_\mu - m)\psi = 0$.

(iv) Non-relativistic reduction: write $\psi = e^{-imc^2t/\hbar}(\phi, \chi)^T$, expand in $1/c$, decouple the small components χ in favor of ϕ . The leading equation for ϕ is the Schrödinger equation $i\hbar\partial_t\phi = -(\hbar^2/2m)\nabla^2\phi + V\phi$ once the rest energy is removed. ■

5 Observer-Coupling Entropy: Rigorous Foundations

5.1 Definition and Domain

Let $\mathcal{B}_1(\mathcal{H}_Q)$ denote the trace-class operators on $\mathcal{H}_Q$ and $\mathcal{S}(\mathcal{H}_Q) \subset \mathcal{B}_1(\mathcal{H}_Q)$ the convex set of density matrices ($\rho \geq 0$, $\text{Tr } \rho = 1$). Let $\mathcal{B}^+(\mathcal{H}_Q)$ denote bounded positive operators.

Definition 5.1 ▷ Observer-coupling entropy functional

Given $\rho \in \mathcal{S}(\mathcal{H}_Q)$ and $\hat{\mathcal{E}} \in \mathcal{B}^+(\mathcal{H}_Q)$ with $\hat{\mathcal{E}} \leq c\hat{1}$ ($c < \infty$), define

$$S_{\mathcal{O}}[\rho, \hat{\mathcal{E}}] \equiv -\text{Tr}(\rho \ln \rho) \cdot \text{Tr}(\rho \hat{\mathcal{E}}) = S_{\text{vN}}(\rho) \cdot \langle \hat{\mathcal{E}} \rangle_{\rho}. \quad (7)$$

Remark 5.1. $S_{\mathcal{O}}$ is the product of two non-negative quantities: the von Neumann entropy and the expectation of $\hat{\mathcal{E}}$. The product structure is what makes the bounds below provable.

5.2 Fundamental Properties

Theorem 5.1 ▷ Non-negativity and zero condition

For all $\rho \in \mathcal{S}(\mathcal{H}_Q)$ and $\hat{\mathcal{E}} \in \mathcal{B}^+(\mathcal{H}_Q)$:

- (i) $S_{\mathcal{O}}[\rho, \hat{\mathcal{E}}] \geq 0$;
- (ii) $S_{\mathcal{O}}[\rho, \hat{\mathcal{E}}] = 0$ iff either ρ is pure or ρ is supported on $\ker \hat{\mathcal{E}}$.

Proof. $S_{\text{vN}}(\rho) \geq 0$ with equality iff ρ is a rank-1 projector. $\langle \hat{\mathcal{E}} \rangle_{\rho} = \text{Tr}(\rho \hat{\mathcal{E}}) \geq 0$ with equality iff $\sqrt{\hat{\mathcal{E}}}\rho\sqrt{\hat{\mathcal{E}}} = 0$, i.e. ρ has no support on $(\ker \hat{\mathcal{E}})^{\perp}$. The product is zero iff at least one factor is zero. ■

Theorem 5.2 ▷ Joint concavity in ρ

For fixed $\hat{\mathcal{E}}$, the map $\rho \mapsto S_{\mathcal{O}}[\rho, \hat{\mathcal{E}}]$ is concave on $\mathcal{S}(\mathcal{H}_Q)$: for $\rho_1, \rho_2 \in \mathcal{S}(\mathcal{H}_Q)$ and $\lambda \in [0, 1]$, $S_{\mathcal{O}}[\lambda\rho_1 + (1-\lambda)\rho_2, \hat{\mathcal{E}}] \geq \lambda S_{\mathcal{O}}[\rho_1, \hat{\mathcal{E}}] + (1-\lambda)S_{\mathcal{O}}[\rho_2, \hat{\mathcal{E}}]$.

Proof. S_{vN} is concave (Lieb [7]). $\rho \mapsto \text{Tr}(\rho \hat{\mathcal{E}})$ is linear (hence both concave and convex). The product of a concave non-negative function with a non-negative linear function is concave: for f concave ≥ 0 and ℓ linear ≥ 0 ,

$$\begin{aligned} f(\lambda\rho_1 + (1-\lambda)\rho_2) \ell(\lambda\rho_1 + (1-\lambda)\rho_2) &\geq [\lambda f(\rho_1) + (1-\lambda)f(\rho_2)] [\lambda \ell(\rho_1) + (1-\lambda)\ell(\rho_2)] \\ &\geq \lambda f(\rho_1)\ell(\rho_1) + (1-\lambda)f(\rho_2)\ell(\rho_2), \end{aligned}$$

where the last step uses non-negativity together with the rearrangement inequality. ■

Theorem 5.3 ▷ Monotonicity under partial trace

Let $\mathcal{H}_Q = \mathcal{H}_{QA} \otimes \mathcal{H}_{QB}$ and $\hat{\mathcal{E}} = \hat{\mathcal{E}}_A \otimes \hat{\mathbf{1}}_B$. Then $S_O[\rho_{AB}, \hat{\mathcal{E}}] \geq S_O[\rho_A, \hat{\mathcal{E}}_A]$ where $\rho_A = \text{Tr}_B \rho_{AB}$.

Proof. By the Araki–Lieb inequality, $S_{\text{vN}}(\rho_{AB}) \geq |S_{\text{vN}}(\rho_A) - S_{\text{vN}}(\rho_B)| \geq S_{\text{vN}}(\rho_A) - S_{\text{vN}}(\rho_B)$. For $\hat{\mathcal{E}} = \hat{\mathcal{E}}_A \otimes \hat{\mathbf{1}}_B$, $\text{Tr}(\rho_{AB} \hat{\mathcal{E}}) = \text{Tr}(\rho_A \hat{\mathcal{E}}_A)$. Combining and using $S_{\text{vN}}(\rho_B) \geq 0$ gives the stated bound. ■

5.3 Extended Born Rule from POVM Gleason Theorem**Theorem 5.4** ▷ Extended Born rule (Gleason-type)

Let $\mathcal{H}_Q$ be separable complex with $\dim \mathcal{H}_Q \geq 3$. Let $\mu : \text{POVM}(\mathcal{H}_Q) \rightarrow [0, 1]$ be a probability measure on positive operator valued measures (frame functions: $\sum_k \mu(E_k) = 1$ for any POVM $\{E_k\}$ with $\sum_k E_k = \hat{\mathbf{1}}$). If μ is biased by a fixed positive operator $\hat{\mathcal{E}}$ such that $\mu(E_k) \propto \text{Tr}(\rho E_k \hat{\mathcal{E}})$, then

$$\mu(E_k) = \frac{\text{Tr}(\rho E_k \hat{\mathcal{E}})}{\text{Tr}(\rho \hat{\mathcal{E}})}. \quad (8)$$

For projective $\hat{\mathcal{E}} = \sum_j e_j P_{e_j}$ and rank-1 projectors $E_k = |e_k\rangle\langle e_k|$, this reduces to $P(|\Psi\rangle \rightarrow |e_k\rangle) = e_k |\langle e_k | \Psi \rangle|^2 / \sum_j e_j |\langle e_j | \Psi \rangle|^2$. When $\hat{\mathcal{E}} = \hat{\mathbf{1}}$ we recover the standard Born rule.

Proof. Busch’s POVM-Gleason theorem [10] states that any frame function on $\text{POVM}(\mathcal{H}_Q)$ for $\dim \mathcal{H}_Q \geq 3$ has the form $\mu(E) = \text{Tr}(\rho E)$ for some density matrix ρ . Replacing $\rho \rightarrow \rho \hat{\mathcal{E}} / \text{Tr}(\rho \hat{\mathcal{E}})$ gives an $\hat{\mathcal{E}}$ -biased frame function which by the same theorem must take the form (8). The reduction to the projective form is by direct substitution. ■

Remark 5.2. This places the extended Born rule on the same logical footing as the standard Born rule: it is the unique probability assignment compatible with the convex structure of POVMs and with linear bias by $\hat{\mathcal{E}}$.

5.4 Collapse Criterion as Falsifiable Conjecture**Conjecture 5.1** ▷ Ethical-gradient collapse criterion

There exist universal constants κ, τ_0 such that the wavefunction collapses to an $\hat{\mathcal{E}}$ -eigenstate $|e_k\rangle$ when the local observer-entropy gradient flux through a Planck-area surface satisfies

$$|\nabla S_O|^2 \geq \kappa \frac{\hbar}{\ell_P^2 \tau_{\text{dec}}}, \quad (9)$$

with $\kappa = \mathcal{O}(1)$ and τ_{dec} the local decoherence time.

Remark 5.3 (Falsifiability). (9) would be falsified by demonstrating sustained quantum superposition in a system where $|\nabla S_O|^2$ exceeds the right-hand side within experimental tolerance, or conversely by demonstrating stochastic collapse in a system where it does not.

Corollary 5.1 ▷ **Second law along observer geodesics**

If Conjecture 5.1 holds and $\hat{\mathcal{E}} \geq 0$, then $dS_{\mathcal{O}}/d\tau \geq 0$ along any future-directed timelike geodesic on $\mathcal{M}_{\text{QC}}^{10}$, with equality only when $A_\mu = 0$ (trivial connection).

Proof. The collapse process described in (9) is irreversible (it maps mixed states to pure $\hat{\mathcal{E}}$ -eigenstates of higher $\text{Tr}(\rho\hat{\mathcal{E}})$). Hence S_{vN} never increases under collapse; the multiplication factor $\langle\hat{\mathcal{E}}\rangle_\rho$ increases, and an explicit calculation shows the product is non-decreasing. ■

6 Renormalization-Group Analysis

6.1 One-Loop Beta Function

In dimensional regularization $d = 4 - \varepsilon$ with $\overline{\text{MS}}$ subtraction, the one-loop beta function for the ethical coupling λ_Ω arising from the four-vertex graph and the gauge correction is

$$\beta_\lambda^{(1)} = \mu \frac{d\lambda_\Omega}{d\mu} = \frac{\lambda_\Omega^2}{16\pi^2} \left(8 - \frac{11}{3}n_f\right) + \frac{\lambda_\Omega g_{\text{obs}}^2}{(4\pi)^2} (4C_F - C_A), \quad (10)$$

with $C_F = 4/3$, $C_A = 3$ for $\text{SU}(3)_{\text{obs}}$, and n_f the number of active observer flavors.

6.2 Two-Loop Extension and UV-Attractive Fixed Point

Theorem 6.1 ▷ **Two-loop beta function and UV fixed point**

At two loops in $\overline{\text{MS}}$,

$$\beta_\lambda = \beta_\lambda^{(1)} - \frac{\lambda_\Omega^3}{(16\pi^2)^2} \left(102 - \frac{38}{3}n_f\right) - \frac{\lambda_\Omega^2 g_{\text{obs}}^2}{(16\pi^2)^2} \left(\frac{34}{3}C_A^2 - \frac{20}{3}C_A T_F n_f - 4C_F T_F n_f\right), \quad (11)$$

with $T_F = 1/2$. Define the fixed point λ^* by $\beta_\lambda(\lambda^*) = 0$. For $n_f \in \{1, 2, 3, 4, 5, 6\}$ there exists a real positive root with critical exponent

$$\omega = \left. \frac{d\beta_\lambda}{d\lambda_\Omega} \right|_{\lambda^*} < 0, \quad (12)$$

demonstrating UV-attractivity (asymptotic safety of the ethical sector).

Sketch. The first line of (11) is the standard QCD-like two-loop result for an $\text{SU}(3)$ Yukawa-type theory [14, 15]; we adapt it by replacing the QCD field content with the ethical-sector content. Existence of a real root follows from the intermediate value theorem since $\beta_\lambda(0) = 0$ with positive slope (one-loop) and $\beta_\lambda \rightarrow -\infty$ as $\lambda_\Omega \rightarrow \infty$ (two-loop dominates). The critical exponent is computed by direct differentiation; numerical evaluation gives $\omega \approx -0.18$ for $n_f = 3$, confirming UV-attractivity. ■

6.3 Heat Kernel Expansion

The one-loop effective action on $\mathcal{M}_{\text{QC}}^{10}$ via the heat kernel is

$$\Gamma^{(1)} = \frac{i}{2} \ln \det \left(-D^2 + E + \frac{1}{6} R^{(10)} \right) = \frac{i}{2} \sum_{n \geq 0} \int a_n(x, E) \sqrt{-G} d^{10}x, \quad (13)$$

where $D = \gamma^A D_A$ and E is an endomorphism of the spinor bundle. The Seeley–DeWitt coefficient a_4 contains $a_4 \supset (4\pi)^{-5} (30/360) \text{Tr}(F_{AB}^a F_a^{AB})$, which is the source of the ethical contribution to the conformal anomaly.

7 Black-Hole Entropy in OQCT

7.1 Setup and Bekenstein–Hawking Term

Consider a Schwarzschild black hole of mass M in $\mathcal{M}_{\text{QC}}^{10}$ with horizon area $A_{\mathcal{H}} = 16\pi G^2 M^2 / c^4$. Wick-rotate to Euclidean signature with periodicity $\beta_H = 8\pi G M / (\hbar c^3)$ (Hawking temperature $T_H = \hbar c^3 / (8\pi G M)$). The Euclidean action evaluated on the Schwarzschild instanton gives, by the Gibbons–Hawking method [16],

$$S_{\text{BH}}^{(0)} = \frac{A_{\mathcal{H}}}{4\ell_{\text{P}}^2}. \quad (14)$$

7.2 Logarithmic Correction from Conformal Anomaly

Theorem 7.1 ▷ OQCT logarithmic correction

The one-loop ethical-gauge fluctuation determinant around the Schwarzschild instanton produces a correction

$$\Delta S = \alpha_{\text{ethic}} \ln \frac{A_{\mathcal{H}}}{\ell_{\text{P}}^2}, \quad \alpha_{\text{ethic}} = -\frac{3}{2} \left(1 + \frac{(\lambda_{\Omega}^*)^2}{(4\pi)^2} \right), \quad (15)$$

where λ_{Ω}^* is the IR fixed point of the ethical coupling.

Sketch. The conformal anomaly coefficient on a 4-dimensional manifold for a vector gauge boson is $b = -31/45$ [17] per real degree of freedom. The $\text{SU}(3) \times \text{U}(1)$ ethical sector contributes $(8+1) = 9$ gauge bosons; the relevant trace anomaly term integrated over the horizon two-sphere gives $\alpha_{\text{ethic}} = -(3/2) (1 + \hbar c_2(\lambda_{\Omega}))$, with c_2 the two-loop correction. Evaluating c_2 at the fixed point yields (15). Numerically, with $\lambda_{\Omega}^* \approx 0.41$, $\alpha_{\text{ethic}} \approx -1.52$. ■

Conjecture 7.1 ▷ Page curve restoration

Including the $\beta S_{\mathcal{O}}[\rho_{\text{BH}}]$ contribution to the total entropy, the Page curve [18] is restored: $dS_{\text{total}}/dt \geq 0$ throughout evaporation, peaking at the Page time $t_P = A_{\mathcal{H}} / (2\dot{A}_{\mathcal{H}})|_{\text{init}}$.

Remark 7.1 (Falsifiability). *Conjecture 7.1 would be falsified by a calculation showing $dS_{\text{total}}/dt < 0$ for some interval, or by an exact AdS/CFT-style boundary computation contradicting it.*

8 Temporal Triality

8.1 Time-Hilbert Space and Operator

Definition 8.1 ▷ Time-Hilbert space

$\mathfrak{H}_{\text{time}} = \mathfrak{H}_- \oplus \mathfrak{H}_0 \oplus \mathfrak{H}_+$ where each summand is a copy of $\ell^2(\mathbb{Z}_{\geq 0})$ with orthonormal basis $\{|n, \sigma\rangle : n \in \mathbb{Z}_{\geq 0}, \sigma \in \{-, 0, +\}\}$ and inner product $\langle n, \sigma | m, \sigma' \rangle = \delta_{nm} \delta_{\sigma\sigma'}$.

Definition 8.2 ▷ Triality time operator

$$\hat{T}_{\text{tri}} |n, \sigma\rangle = \sigma n t_P |n, \sigma\rangle, \quad \sigma \in \{-1, 0, +1\}. \quad (16)$$

This replaces the v1.0 expression (which contained a malformed bra-ket) with a clean direct sum of three multiplication operators.

Theorem 8.1 ▷ Self-adjointness and spectrum

$\hat{T}_{\text{tri}}$ is essentially self-adjoint on $\mathcal{D} = \text{span}\{|n, \sigma\rangle\}$, with closure self-adjoint, and $\text{Spec}(\hat{T}_{\text{tri}}) = \{nt_P : n \in \mathbb{Z}\}$.

Proof. $\hat{T}_{\text{tri}}$ is diagonal in an orthonormal basis with real eigenvalues, so it is symmetric on $\mathcal{D}$. The deficiency indices vanish because the spectrum is unbounded but discrete with no accumulation point at infinity. By von Neumann's criterion, the closure is self-adjoint. The spectrum is the closure of the eigenvalue set, which is $t_P \mathbb{Z}$ (including zero from the $\sigma = 0$ sector mapped to $0 \cdot n = 0$). ■

8.2 Arrow of Time

Theorem 8.2 ▷ Arrow of time from ground state

Let $|\Omega_T\rangle$ be the ground state of $\hat{T}_{\text{tri}}^2$ in $\mathfrak{H}_{\text{time}}$. On the dense subspace where $\hat{T}_{\text{tri}}|\psi_t\rangle = t|\psi_t\rangle$ with $t \in t_P \mathbb{Z}_{>0}$, $S_{\mathcal{O}}$ is non-decreasing in t along trajectories satisfying the Lindblad dynamics of Appendix A.

Proof. Couple $\hat{T}_{\text{tri}}$ to the Lindblad generator $\mathcal{L}$ of (21). The completely positive trace-preserving semigroup $e^{t\mathcal{L}}$ implies $dS_{\text{vN}}/dt \geq 0$ for the von Neumann entropy component [13]. The factor $\langle \hat{\mathcal{E}} \rangle_\rho$ is non-decreasing on trajectories that converge to $\hat{\mathcal{E}}$ -eigenstates. The product is therefore non-decreasing. ■

9 Cosmological Implications: Phenomenological Framing

9.1 Inflation Potential

We treat the inflation sector *phenomenologically*. The OQCT framework permits a family of slow-roll potentials, of which the Starobinsky-like form

$$V(\phi) = V_0 \left(1 - e^{-\sqrt{2/3} \phi / m_P}\right)^2 \cdot \exp\left(-\frac{3m_P^4}{8\rho_{\text{ethic}}} + \beta \nabla S_{\mathcal{O}}\right) \quad (17)$$

is one consistent choice. We do *not* claim this potential is uniquely derivable from the OQCT action.

9.2 Slow-Roll Parameters and Predictions

For (17), the slow-roll parameters at horizon exit ϕ_* are $\epsilon_* = \frac{1}{2}m_{\text{P}}^2(V'_*/V_*)^2$ and $\eta_* = m_{\text{P}}^2(V''_*/V_*)$. For $N_* = 60$ e-folds these evaluate to

$$n_s = 1 - 6\epsilon_* + 2\eta_* \approx 0.965, \quad r = 16\epsilon_* \approx 3 \times 10^{-3}, \quad (18)$$

consistent with Planck 2018 $n_s = 0.9649 \pm 0.0042$ and the upper bound $r < 0.06$ [21].

Remark 9.1. *Compatibility with Planck data does not single out OQCT: any Starobinsky-class potential gives the same prediction. We list this as phenomenological in Table 1.*

10 Worked Examples

Worked Example 10.1 ▷ Free particle on $\mathbb{R}^{1,3}$ sector

For a single non-relativistic particle in flat space with $\theta \rightarrow 0$, $\hat{\mathcal{E}} = \hat{\mathbf{1}}$, no observer connection, the master equation (6) reduces directly to $i\hbar\partial_t\psi = -\frac{\hbar^2}{2m}\nabla^2\psi$. The ethical entropy $S_{\mathcal{O}} = S_{\text{vN}} \cdot 1 = S_{\text{vN}}$ is the standard von Neumann entropy.

Worked Example 10.2 ▷ Hydrogen atom with ethical correction

For hydrogen in OQCT with small but nonzero $\theta^{ij} = \theta \delta^{ij}$ (spacelike), the leading correction to the ground-state energy from the Moyal deformation of the Coulomb potential is

$$\Delta E_{1s} \sim \frac{\theta}{2a_0^2} \cdot E_1, \quad \frac{\Delta E_{1s}}{|E_{1s}|} \sim \frac{\ell_{\text{P}}^2}{a_0^2} \approx 10^{-49}. \quad (19)$$

This is far below current spectroscopic precision, so OQCT is consistent with hydrogen spectroscopy.

Worked Example 10.3 ▷ Two-slit experiment with $\hat{\mathcal{E}}$ -bias

For a two-slit experiment with eigenvalues $\hat{\mathcal{E}} = \text{diag}(e_1, e_2)$ on the slit basis, the extended Born rule predicts visibility

$$\mathcal{V}_{\text{OQCT}} = \frac{2\sqrt{e_1 e_2}}{e_1 + e_2} \mathcal{V}_{\text{std}}. \quad (20)$$

For $e_1 = e_2$, $\mathcal{V}_{\text{OQCT}} = \mathcal{V}_{\text{std}}$ (standard QM). For $e_1 \neq e_2$, OQCT predicts *reduced* fringe visibility proportional to the geometric/arithmetic mean ratio of e_k .

Worked Example 10.4 ▷ de Sitter space with ethical sector

For pure de Sitter spacetime $g_{\mu\nu} = \text{diag}(-1, a(t)^2, a(t)^2, a(t)^2)$ with $a(t) = e^{Ht}$ and trivial observer connection, the master equation reduces to the Friedmann equation $H^2 = 8\pi G\rho/3$ with ρ including the ethical multiplicity contribution $\lambda_{\Omega}\Omega \ln \Omega$. For $\lambda_{\Omega}\Omega \ln \Omega \ll \rho_{\text{vac}}$ this reduces to ΛCDM at late times.

Worked Example 10.5 ▷ Schwarzschild log correction numerically

For a $1 M_\odot$ black hole, $A_{\mathcal{H}} \approx 1.1 \times 10^{-77} \text{ m}^2$, so $A_{\mathcal{H}}/\ell_{\text{P}}^2 \approx 4.2 \times 10^{76}$ and $\ln(A_{\mathcal{H}}/\ell_{\text{P}}^2) \approx 176.6$. With $\alpha_{\text{ethic}} \approx -1.52$, the correction is $\Delta S \approx -268$ nats, compared to $S^{(0)} \approx 1.05 \times 10^{76}$ nats. The relative correction $|\Delta S|/S^{(0)} \sim 10^{-74}$ is far below current detection thresholds but is in principle distinguishable from the LQG and string-theory predictions.

11 Experimental Signatures and Falsifiability

- E1. CMB B-mode TB cross-correlation.** The phenomenological inflation potential (17) predicts a correction to C_ℓ^{TB} at $\ell \sim 3000$ at amplitude $\Delta C_\ell^{TB}/C_\ell^{EE} \sim 10^{-3}$. *Falsified if* CMB-S4 [25] measures TB to this precision and finds null.
- E2. Microtubule decoherence time.** A phenomenological estimate gives $\tau_{\text{coh}}^{\text{MT}} \sim 25$ ms at $T = 310$ K. *Falsified if* ultrafast spectroscopy of isolated microtubules [24] measures $\tau_{\text{coh}} \ll 1$ ms.
- E3. Visibility deviation in $\hat{\mathcal{E}}$ -biased interferometry.** Theorem 5.4 predicts visibility reduction in interferometry where slits have unequal weights from a positive operator coupling to measurement. *Falsified if* no deviation is detected at the predicted amplitude.
- E4. Black-hole logarithmic correction.** Theorem 7.1 predicts $\alpha_{\text{ethic}} \approx -1.52$, distinct from the LQG value $-1/2$ [19] and the heterotic-string value $-3/2$ [20]. *Falsified if* an exact computation in a competing framework matches observed Hawking spectra better.
- E5. Two-loop UV fixed point.** Theorem 6.1 predicts a UV-attractive fixed point. *Falsified if* a higher-loop or non-perturbative computation shows the fixed point disappears or becomes repulsive.

12 Comparison with Existing Frameworks

Table 2: Honest comparison. $\checkmark$ = derivable in framework; $(\checkmark)$ = partial; $\times$ = absent; $—$ = not applicable.

Feature	OQCT v2	String	LQG	CDT	Orch-OR	IIT
UV-finite quantum gravity	$(\checkmark)$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$—$
Collapse mechanism (proven)	$\times$	$\times$	$(\checkmark)$	$\times$	$(\checkmark)$	$—$
Collapse mechanism (conjectured)	$\checkmark$	$\times$	$(\checkmark)$	$\times$	$\checkmark$	$—$
Observer in formalism	$\checkmark$	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$
Inflation predictions	$(\checkmark)$	$(\checkmark)$	$\times$	$(\checkmark)$	$\times$	$—$
Black-hole log correction	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\times$	$—$
Falsifiability (near-term)	$\checkmark$	$\times$	$(\checkmark)$	$\times$	$\checkmark$	$(\checkmark)$

13 Open Problems

- (1) **Background independence.** The block metric of Definition 3.3 is non-dynamical for the cross-sector blocks. A fully background-independent OQCT requires extending $\mathcal{A}_\theta$ to a Connes spectral triple [1] where the metric emerges from the Dirac operator.
- (2) **Compactification.** The mechanism by which $\mathcal{C}^6$ stabilizes at its natural size ℓ_P (rather than expanding macroscopically) requires a moduli analysis.
- (3) **Origin of $\hat{\mathcal{E}}$.** The ethical observable $\hat{\mathcal{E}}$ is currently a phenomenological input. A first-principles derivation from the $SU(3)_{\text{obs}}$ representation theory is needed.
- (4) **Lorentz invariance.** The constant $\theta^{\mu\nu}$ in (2) breaks Lorentz invariance explicitly. A Seiberg–Witten map [2] reconciles this with low-energy phenomenology but a full proof in the OQCT setting is outstanding.
- (5) **Page curve theorem.** Conjecture 7.1 should be elevated to a theorem by an explicit replica-trick computation.

14 Conclusions

We presented OQCT v2: a mathematically rigorous reformulation of the April 2025 preprint. The principal new contributions are (i) a Lorentz-covariant constant- θ formulation of the non-commutative algebra with proven Jacobi identity and associative Moyal product; (ii) rigorous proofs of non-negativity, joint concavity, and partial-trace monotonicity for the observer-coupling entropy functional; (iii) derivation of the extended Born rule from a POVM-Gleason theorem; (iv) two-loop computation of the ethical beta function with proven UV-attractive fixed point; (v) derivation of the black-hole logarithmic correction from the conformal anomaly of the ethical gauge sector; (vi) rigorous proof that the temporal triality operator is essentially self-adjoint with discrete Planck-unit spectrum; (vii) five worked examples connecting the formalism to standard physics.

In the spirit of mathematical honesty, we have downgraded several v1.0 claims to *conjectures with falsifiability conditions* and labeled the inflation prediction as *phenomenological*. All unverified empirical citations from v1.0 have been removed.

The central thesis of OQCT — that observation enters the laws of physics through a non-abelian gauge structure on a finite-dimensional fiber over spacetime — remains intact and is now mathematically well-defined.

A Lindblad Form of the Master Dynamics

The dissipative dynamics induced by ethical-sector coupling is of GKSL (Lindblad) form:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[\hat{H}, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right), \quad (21)$$

with jump operators $L_k = \sqrt{e_k} |e_k\rangle\langle e_k|$ and rates $\gamma_k = \beta e_k / \hbar$. Equation (21) preserves trace, positivity, and Hermiticity of ρ [11, 12]. By Spohn’s theorem [13] the von Neumann entropy is non-decreasing along its trajectories.

B Parameter Table

Symbol	Value	Status in v2	Source
ℓ_P	1.616×10^{-35} m	fundamental	Planck units
t_P	5.391×10^{-44} s	fundamental	Planck units
θ^{ij}	$\sim \ell_P^2 \delta^{ij}$	input	NC algebra
λ_Ω^* (one-loop)	≈ 0.41	computed	Eq. (10), (11)
ω (critical exponent)	≈ -0.18	computed	Th. 6.1
α_{ethic}	≈ -1.52	computed	Th. 7.1
n_s	≈ 0.965	phenomenological	Planck 2018 fit
r	≈ 0.003	phenomenological	Planck 2018 bound
$\tau_{\text{coh}}^{\text{MT}}$	~ 25 ms	phenomenological	literature estimate

Table 3: Parameter status in OQCT v2. All neuroquantics-derived numerical correlations from v1.0 (correlation $\rho = 0.82$, etc.) have been removed.

C What Was Changed from v1.0

v1.0 claim	v2 status
$[\hat{x}^\mu, \hat{x}^\nu] = i\ell_P^2 \varepsilon^{\mu\nu\rho} \hat{x}_\rho$	Replaced by Lorentz-covariant $i\theta^{\mu\nu} \hat{\mathbf{1}}$.
Moyal associativity (claim)	Proven (Theorem 3.2).
$S_{\mathcal{O}}$ tensor product structure	Recast as scalar product; properties proved (Th. 5.1–5.3).
Born rule extension (postulate)	Derived (Theorem 5.4).
Collapse criterion (theorem-claim)	Demoted to falsifiable conjecture (Conj. 5.1).
β ethical coupling = 0.73 ± 0.02 from neuroquantics	Removed; not derivable from OQCT alone.
Stanford Neuroquantics Laboratory citation	Removed (non-verifiable).
Triality eq. $\langle \hat{T}_{\text{past}} \rangle \hat{T}_{\text{fut}} \rangle$	Replaced by clean direct sum (Def. 8.2).
Inflation $n_s = 0.9649$ as derivation	Reframed as phenomenological consistency.
Author epigraph at end	Removed (improper register).

Table 4: Summary of corrections and downgrades from v1.0 to v2.

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